

LINEAR AND HIERARCHICAL LINEAR MODELS

Mixed Models, and Multilevel Models, GLMMs, etc...

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TODAY

- Linear Models
- Short break 😊
- What are hierarchical linear models
- Learn how to build and interpret linear mixed models
- Break 😊
- Practice



REQUIRED MATERIAL

<https://github.com/lacapary/BIO503/>

REQUIRED MATERIAL

You are required to have downloaded and installed

```
1  install.packages (c ("lme4",  
2                        "ggeffects",  
3                        "MuMIn",  
4                        "emmeans"  
5                        ) )
```

REQUIRED MATERIAL

DO NOT HESITATE TO ASK QUESTIONS!

DATA FOR THIS PART OF THE SESSION



Coding Club Mixed models

Used to build intuition for linear models and mixed models.

LINEAR MODELS

WHAT IS A LINEAR MODEL?

In ecology, we often want to ask:

- Does temperature drive species richness?
- Does soil nitrogen limit plant biomass?
- Does drought reduce tree growth?

A **linear model** is our tool for quantifying these relationships formally. It describes how a *response variable* (what we measure) changes as a predictable function of one or more *predictors* (drivers, treatments, environmental gradients), while accounting for unexplained noise in the system.

WHAT IS A LINEAR MODEL?

In its simplest form:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- Y = the biological response you measure (e.g., plant biomass)
- X = the driver or predictor (e.g., soil nitrogen)
- β_0 = baseline value when $X = 0$ (intercept)
- β_1 = how much Y changes per unit of X (the effect size)
- ε = everything else: noise, unmeasured factors, natural variation

WHY DO WE USE LINEAR MODELS?

Ecological systems are *complex*, but linear models give us a *simplified* framework to ask and answer key ecological questions.

-  Simple and interpretable: Easy to understand and communicate.
-  Flexible: Can be extended with multiple predictors, interactions, etc.
-  Hypothesis testing: Test relationships grounded in ecological theory.
-  Data-driven: Connect real-world measurements to ecological mechanisms.
-  Foundation for advanced models: GLMs, mixed models, and beyond.

LINEAR MODELS

A simple example:

Does soil nitrogen content influence plant productivity?

$$\text{Plant biomass} = \beta_0 + \beta_1 \cdot \text{Soil nitrogen} + \varepsilon$$

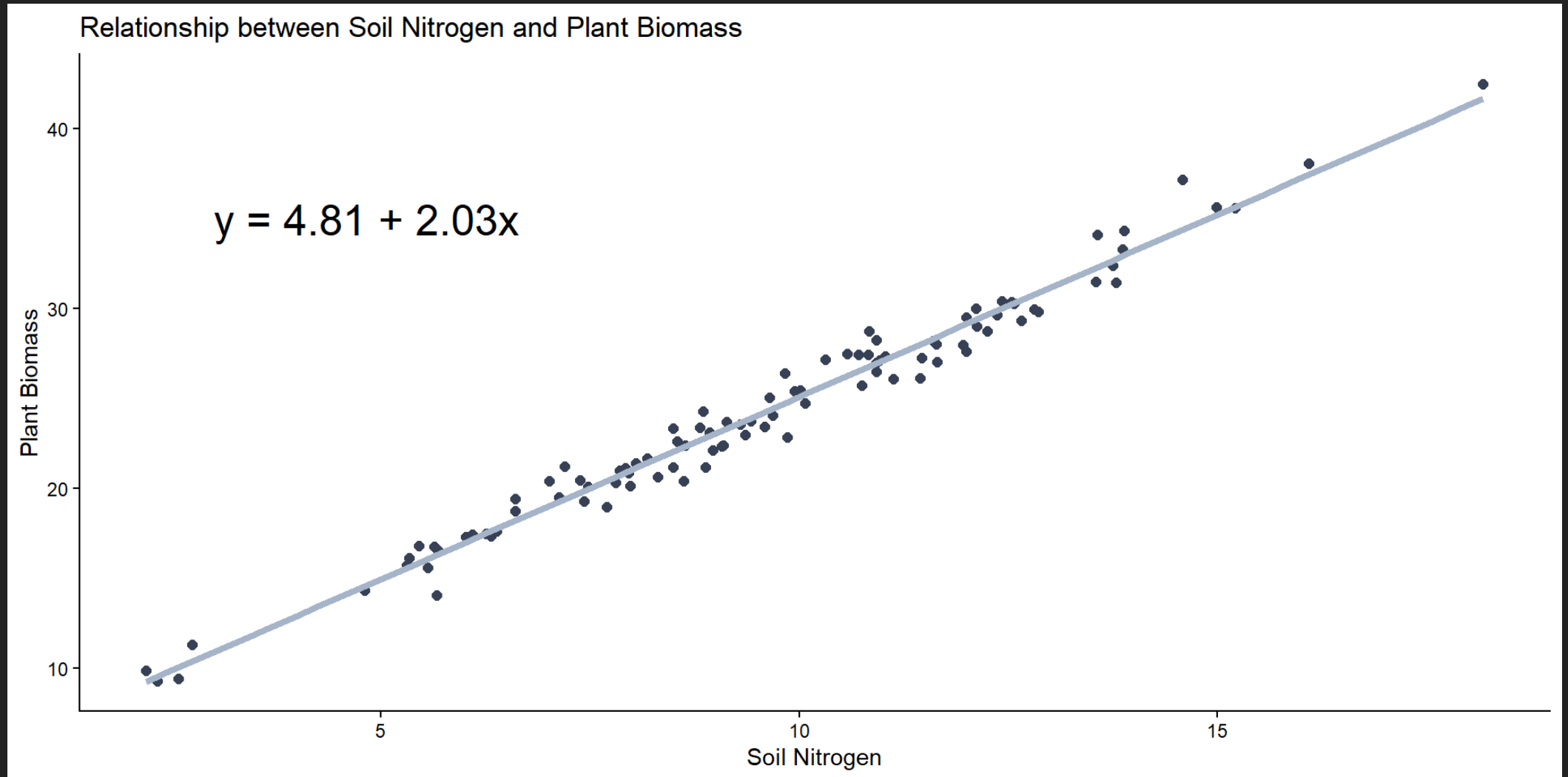
LINEAR MODELS

A simple example:

Does soil nitrogen content influence plant productivity?

$$\text{Plant biomass} = \beta_0 + \beta_1 \cdot \text{Soil nitrogen} + \varepsilon$$

LINEAR MODELS



LINEAR MODELS:

Dragon Example

EXPLORE THE DATA

```
1 library(tidyverse)
2 load("CC-Linear-mixed-models/dragons.RData")
3
4 dragons_Bavarian <- dragons |>
5   filter(mountainRange == "Bavarian")
```

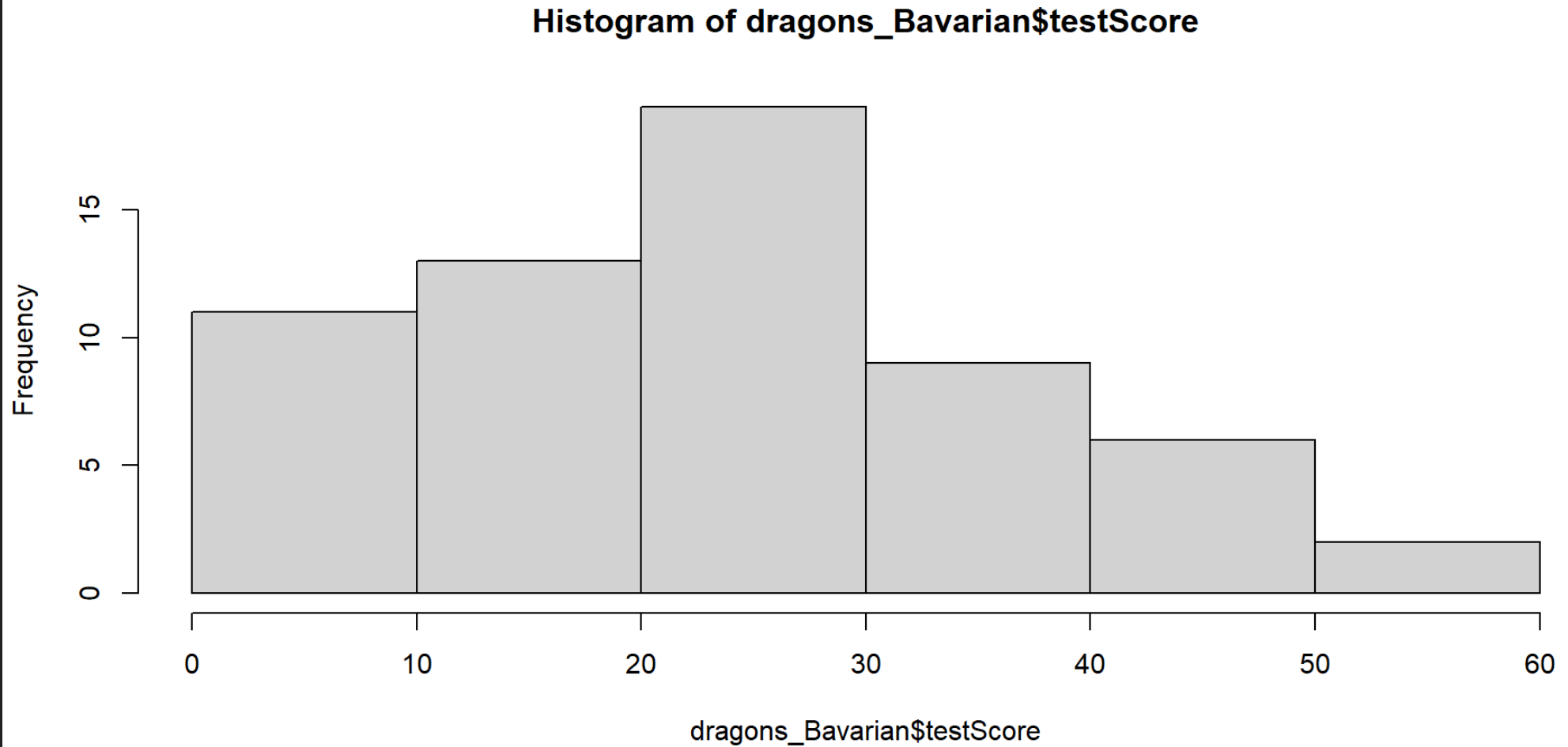
EXPLORE THE DATA

```
1 head(drakens_Bavarian)
```

	testScore	bodyLength	mountainRange	site
1	16.147309	165.5485	Bavarian	a
2	33.886183	167.5593	Bavarian	a
3	6.038333	165.8830	Bavarian	a
4	18.838821	167.6855	Bavarian	a
5	33.862328	169.9597	Bavarian	a
6	47.043246	168.6887	Bavarian	a

EXPLORE THE DATA

```
1 hist(drakens_Bavarian$testScore)
```



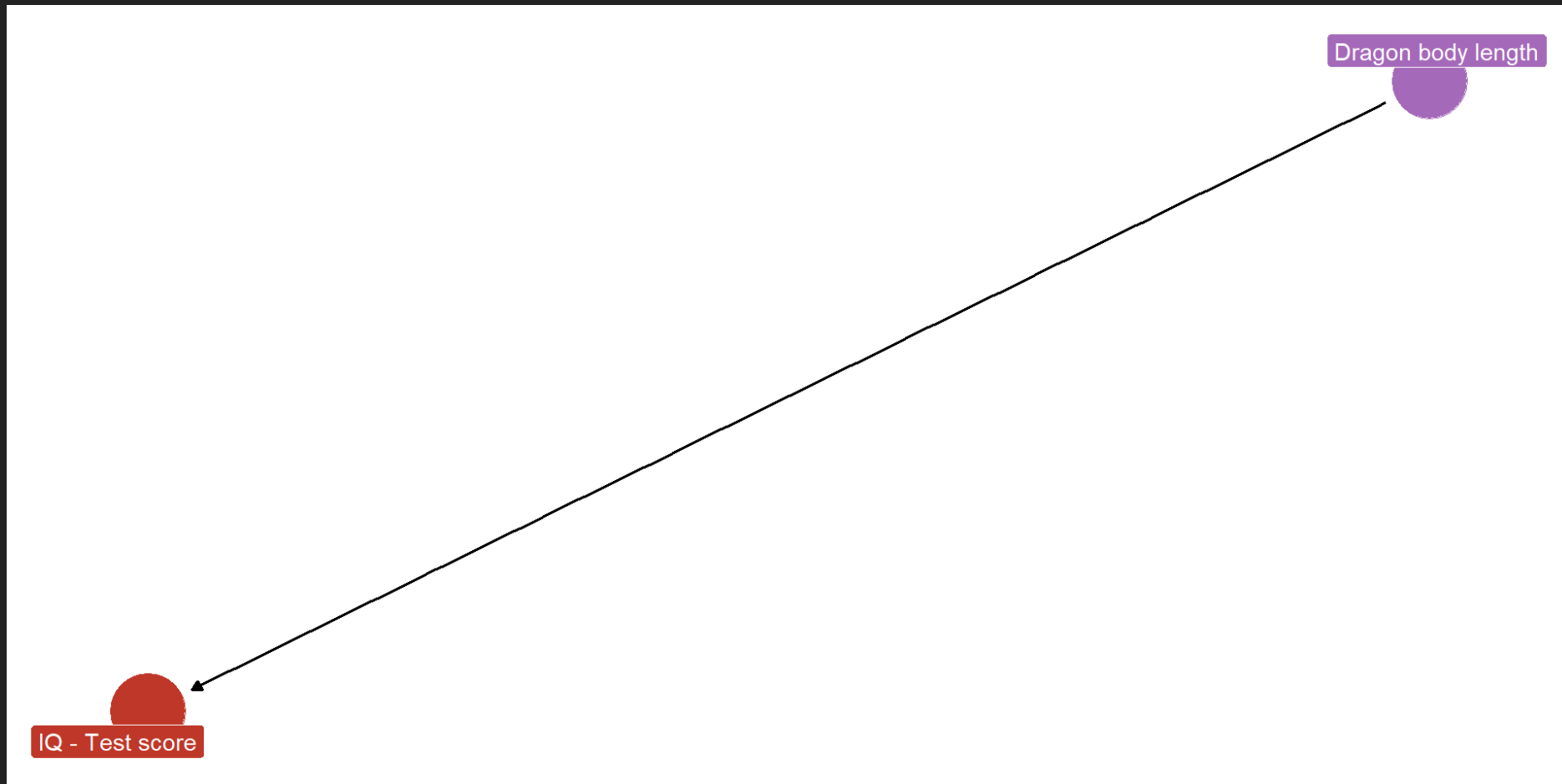
OUR QUESTION

How does body size of dragons influence their intelligence?

OUR QUESTION

How does body size of dragons influence their intelligence?

First we will plot our DAG:



OUR QUESTION

How does body size of dragons influence their intelligence?

Then our model:

$$\text{Test score} = \beta_0 + \beta_1 \cdot \text{Body length} + \varepsilon$$

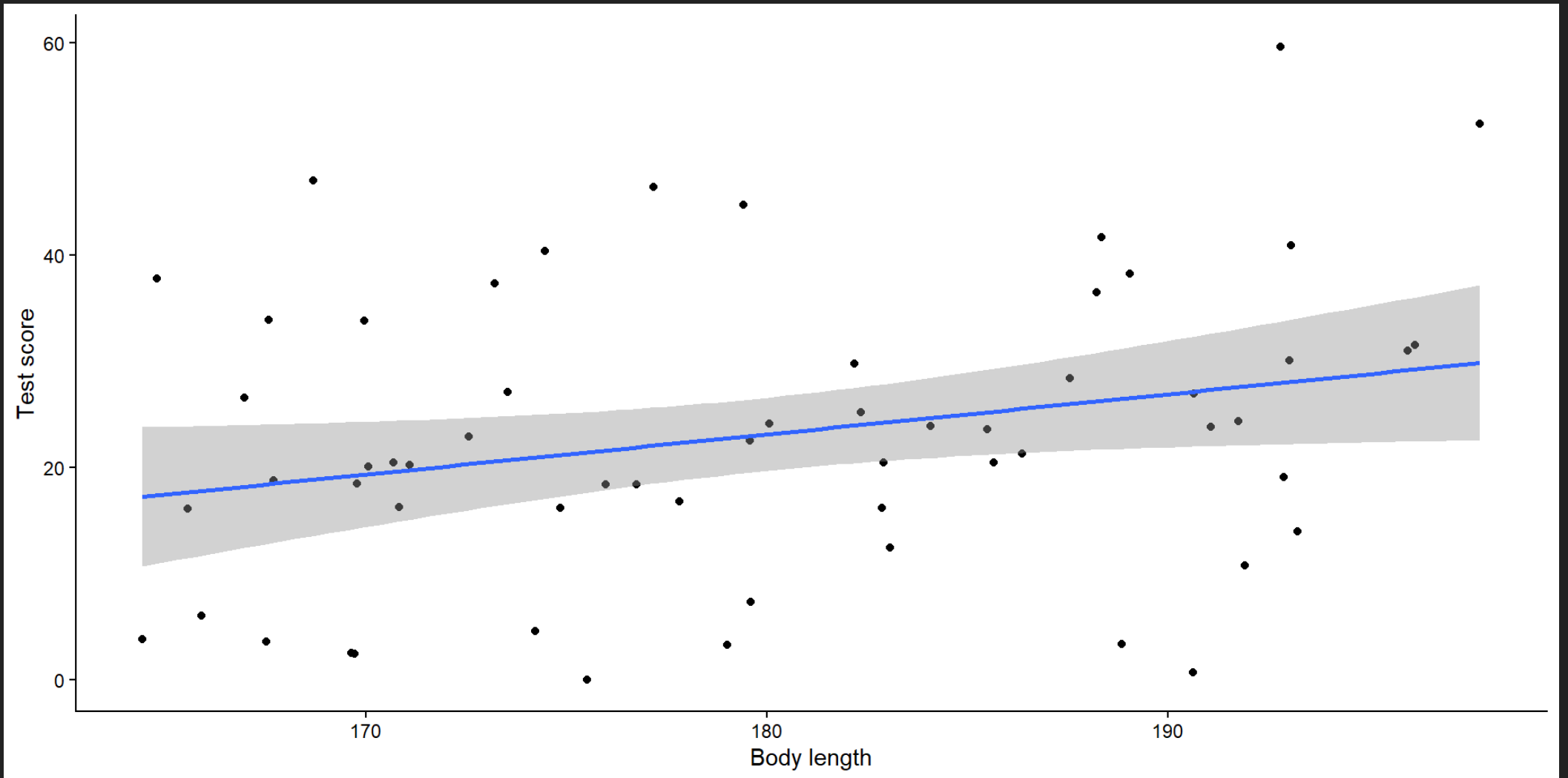
OUR QUESTION

Now lets visualize the relationship using ggplot.

```
1 p <- dragons_Bavarian %>%
2   ggplot(aes(
3     x = bodyLength,
4     y = testScore)) +
5   geom_point() +
6   stat_smooth(method = "lm") +
7   theme_classic() +
8   labs(
9     x = "Body length",
10    y = "Test score")
```

OUR QUESTION

Now lets visualize the relationship using ggplot.



LETS MODEL IT

```
1 basic.lm <- lm(testScore ~ bodyLength, data = dragons_Ba
2 summary(basic.lm)
```

Call:

```
lm(formula = testScore ~ bodyLength, data = dragons_Bavarian)
```

Residuals:

Min	1Q	Median	3Q	Max
-26.433	-8.228	-0.594	6.893	31.637

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-44.8343	32.3344	-1.387	0.1709
bodyLength	0.3777	0.1795	2.104	0.0397 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.35 on 58 degrees of freedom

Multiple R-squared: 0.07093, Adjusted R-squared: 0.05491

F-statistic: 4.428 on 1 and 58 DF, p-value: 0.0397

FACTORIAL ANOVA

FACTORIAL ANOVA: IT'S JUST A LINEAR MODEL

- *Factorial ANOVA* is a special kind of analysis used when you have two or more categorical predictors.
- But in R, you don't need a special ANOVA function. You can just use the `lm()` function.
- That's because factorial ANOVA is just a *linear model* with categorical predictors.

FACTORIAL ANOVA: EXAMPLE

Using the `iris` data set.

This fits a linear model where Species is treated as a factor.

```
1 model <- lm(Sepal.Length ~ Species, data = iris)
2 summary(model)
```

Call:

```
lm(formula = Sepal.Length ~ Species, data = iris)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.6880	-0.3285	-0.0060	0.3120	1.3120

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.0060	0.0728	68.762	< 2e-16 ***
Speciesversicolor	0.9300	0.1030	9.033	8.77e-16 ***
Speciesvirginica	1.5820	0.1030	15.366	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5148 on 147 degrees of freedom

Multiple R-squared: 0.6187, Adjusted R-squared: 0.6135

F-statistic: 119.3 on 2 and 147 DF, p-value: < 2.2e-16

FACTORIAL ANOVA: EXAMPLE

When we run the ANOVA, we're checking if the average sepal length is about the same for all the species, or if at least one species is different from the others.

```
1 anova(model)
```

Analysis of Variance Table

Response: Sepal.Length

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Species	2	63.212	31.606	119.26	< 2.2e-16 ***
Residuals	147	38.956	0.265		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

FACTORIAL ANOVA: POST-HOC ANALYSIS

- Let's say we run an ANOVA and find a significant result.
- The test: "There's a difference somewhere"
- The post-hoc test: "Who's different from who"

```
1 library(emmeans)
2 # Get estimated marginal means
3 em <- emmeans(model, ~ Species)
4
5 # Pairwise comparisons with Tukey adjustment
6 pairs(em, adjust = "tukey")
```

contrast	estimate	SE	df	t.ratio	p.value
setosa - versicolor	-0.930	0.103	147	-9.033	<.0001
setosa - virginica	-1.582	0.103	147	-15.366	<.0001
versicolor - virginica	-0.652	0.103	147	-6.333	<.0001

P value adjustment: tukey method for comparing a family of 3 estimates

FACTORIAL ANOVA: POST-HOC ANALYSIS

! Important

Use post-hoc tests only if your ANOVA result is significant.



GENERALISED LINEAR MODELS

FROM LM TO GLM: WHAT CHANGES?

A standard linear model makes three key assumptions:

- The response variable is **continuous**.
- The residuals are **normally distributed**.
- The variance is **constant** across all fitted values (homoscedasticity).

In ecology, these assumptions are routinely violated:

- Counting individuals per plot: integers, never negative, often zero-heavy.
- Presence/absence of a species: binary, bounded between 0 and 1.
- Percentage cover: proportion, can't exceed 100%.

A **Generalised Linear Model (GLM)** relaxes these constraints by letting you choose the distribution that actually fits your data.

FROM LM TO GLM: THE LINK FUNCTION

A GLM has three components:

- **Random component**: the distribution of the response (e.g., Poisson, Binomial).
- **Systematic component**: the linear predictor, $\beta_0 + \beta_1 X + \dots$
- **Link function**: a transformation that connects the linear predictor to the expected value of the response on the correct scale.

POSSIBLE DISTRIBUTIONS

Choose based on the nature of your response variable:

- **Gaussian** — continuous, symmetric data; the classic linear model (e.g., plant height, biomass, temperature).
- **Poisson** — count data; integers, cannot be negative, often zero-heavy (e.g., number of species per plot, insect abundance).
- **Binomial** — binary or proportion data (e.g., presence/absence, proportion of plots occupied by a species).
- **Gamma** — continuous, strictly positive, right-skewed (e.g., biomass, precipitation).
- **Negative Binomial** — count data with overdispersion, when Poisson variance is too small to fit the data.

WHICH MODEL TO USE?

(a non-exhaustive list of...)

COMMON DATA TYPES IN ENVIRONMENTAL SCIENCES AND THEIR ASSOCIATED DISTRIBUTIONS & TESTS

DATA TYPE	DISTRIBUTION: TESTS	EXAMPLES
Continuous data	Gaussian (normal): lm, mixed-effects models	Traits like height, weight, nutrient content, and anything else you can measure across a range centered on a mean value
Count data (whole numbers)	Poisson: glm, glmm	Population counts, number of species in an area, of youngs in a litter, and other discrete measures
Proportion data	If 2 possible outcomes, Binomial: glm, glmm If more outcomes: chi-squared test	Survival, seed germination trials, other events that can be described as “successes” and “failures” Habitat selection (does a species utilise a type of habitat in greater proportion than its availability?), differences in vegetation types between sites or over time

GENERALISED LINEAR MODELS: SYNTAX IN R

Syntax for a model with Poisson distribution (count data)

```
1 glm(y ~ x, family = poisson, data = df)
```

Syntax for a model with Binomial distribution (presence/absence, proportions)

```
1 glm(y ~ x, family = binomial, data = df)
```

BREAK



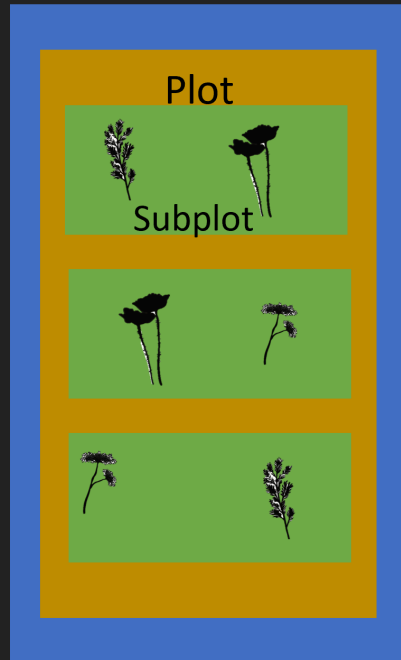
HIERARCHICAL LINEAR MODELS

ECOLOGICAL AND BIOLOGICAL DATA CAN BE COMPLEX!

- Hierarchical structure in the data
- Many covariates and grouping factors
- Unbalanced study/experimental design

HIERARCHICAL STRUCTURES EXAMPLE

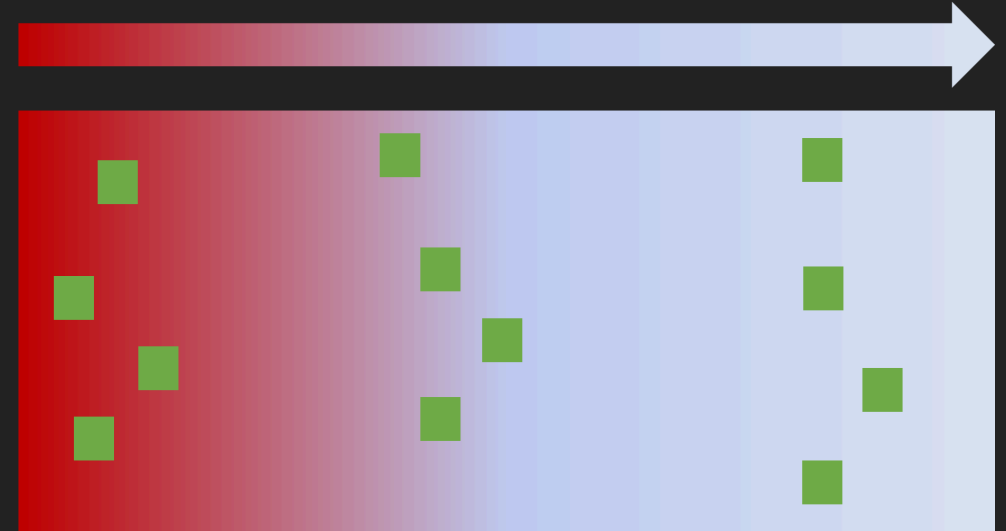
Treatment



Control



Environmental gradient



WHAT IS THE INDEPENDENCE ASSUMPTION?

In basic statistical models, we assume each observation gives us **new, independent information**.

What happens in one sample shouldn't depend on what happened in another.

HIERARCHICAL STRUCTURES IN ECOLOGY

- Measurements from the **same plot or site** are often more similar.
- Observations from the **same species** might behave similarly.
- Samples collected **close in time or space** can be correlated.

ANALOGY

Measuring the same tree's height 3 times isn't the same as measuring 3 different trees.

We call this **dependence** — and we need special models to handle it!

DIAGNOSING DEPENDENCE

Ask yourself:

Are there clusters in my data?

Are data points grouped by site, species, individual, or time?

HOW COULD WE ANALYZE THIS DATA?

A standard linear model treats all observations as independent. But we just established they are not.

If we ignore the grouping structure, we risk:

- **Pseudoreplication**: treating 10 measurements from the same site as 10 independent data points.
- **Wrong p-values**: artificially inflating our confidence in the results.
- **Misleading conclusions**: a pattern that exists only within one site gets presented as a general rule.

HOW COULD WE ANALYZE THIS DATA?

We need a model that can separate two things at once:

1. The **biological effect we care about** (e.g., does temperature reduce species richness?).
2. The **noise introduced by the grouping structure** (e.g., sites differ from each other for all sorts of reasons we did not measure).

That model is a **Linear Mixed Model**.

It handles both by splitting predictors into two types: **fixed effects** and **random effects**.

FIXED EFFECTS

Fixed effects are the predictors you explicitly want to test and report.

- Each level gets its own estimated coefficient in the model.
- They directly answer your research question.
- You have measured all the levels you care about, and you would report them specifically in your results.

Ecological examples: treatment (warmed vs. control), soil nutrient level, land use category, species identity (when comparing specific species is the goal).

RANDOM EFFECTS

Random effects are grouping variables where the specific levels in your study are a sample from a larger population of possible levels.

- The model does not estimate a separate coefficient for each group.
- Instead, it estimates how much variability exists across groups, treating each group's effect as a random draw from a shared distribution.
- You are not asking “what is specific to Site 3?” but rather “how much do sites vary overall, so I can account for it?”

Ecological examples: site, plot, individual plant (repeated measures), year, observer. They are always **categorical**.

ACTIVE BREAK

Scenario: You are studying how warming affects plant height in Arctic tundra. You set up open-top chambers (warming treatment) and control plots across **6 sites** spread along a latitudinal gradient. At each site you have 5 warmed plots and 5 control plots, and you measure the height of 10 plants per plot.

Your research question: Does warming increase plant height?

Discuss with your neighbour:

1. **What is the fixed effect?** What predictor directly answers your research question?
2. **What is the random effect?** The 6 sites were chosen because they were accessible, not because you care about those specific sites. What does that make them?

BACK TO THE DRAGONS:

now with all mountain ranges

DRAGON DATA

We went to the field and sampled more mountains with dragons.

```
# A tibble: 24 × 3
# Groups:   mountainRange [8]
  mountainRange site  samples
  <fct>         <fct>    <int>
1 Bavarian      a         20
2 Bavarian      b         20
3 Bavarian      c         20
4 Central       a         20
5 Central       b         20
6 Central       c         20
7 Emmental      a         20
8 Emmental      b         20
9 Emmental      c         20
10 Julian       a         20
# i 14 more rows
```

RESEARCH QUESTION

IS THE TEST SCORE AFFECTED BY BODY LENGTH?

DRAGONS: NAIVE LINEAR MODEL

```
1 basic.lm <- lm(testScore ~ bodyLength2, data = dragons)
2 summary(basic.lm)
```

Call:

```
lm(formula = testScore ~ bodyLength2, data = dragons)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-56.962	-16.411	-0.783	15.193	55.200

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	50.3860	0.9676	52.072	<2e-16 ***
bodyLength2	8.9956	0.9686	9.287	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

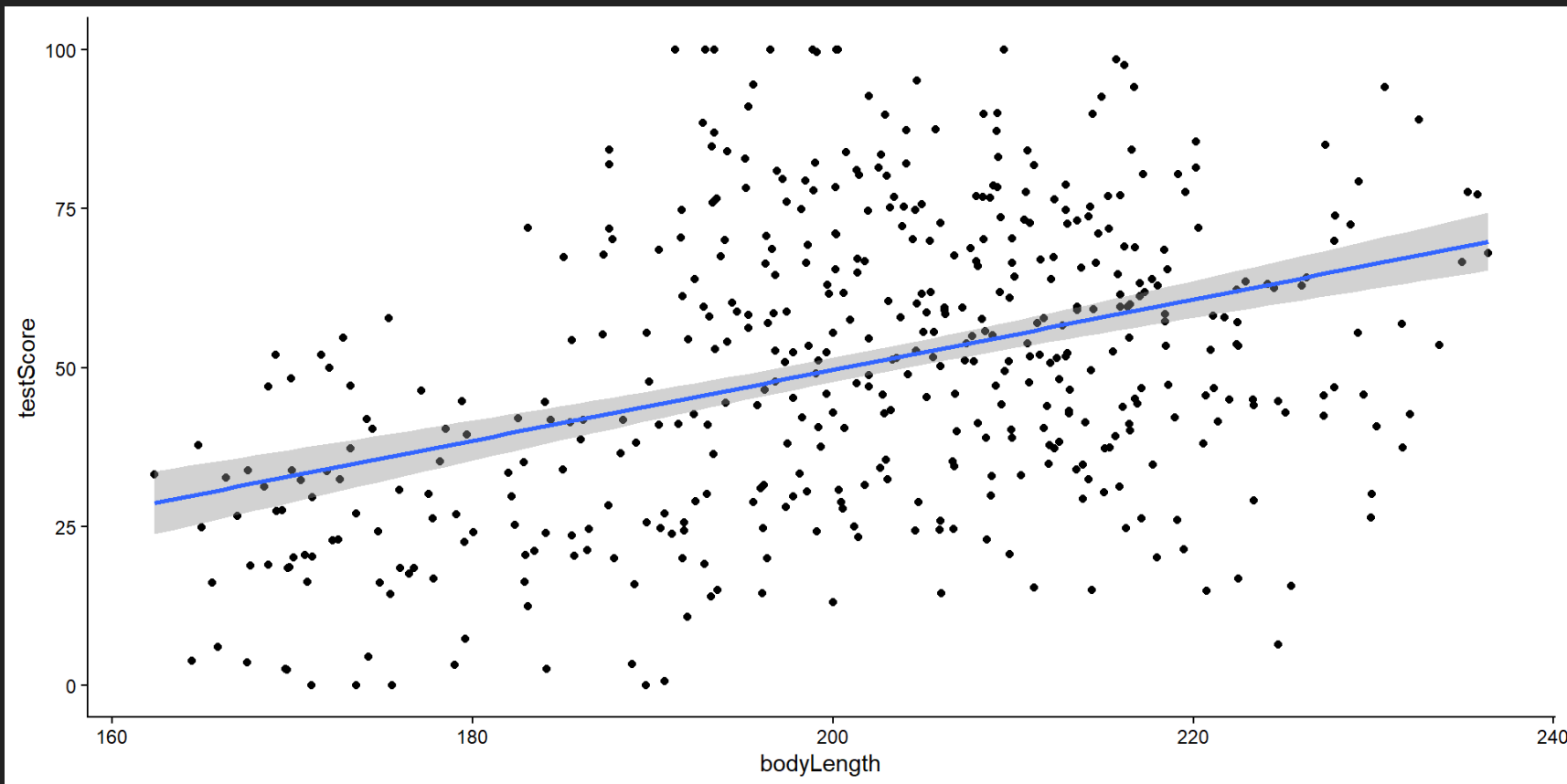
Residual standard error: 21.2 on 478 degrees of freedom

Multiple R-squared: 0.1529, Adjusted R-squared: 0.1511

F-statistic: 86.25 on 1 and 478 DF, p-value: < 2.2e-16

DRAGONS: NAIVE LINEAR MODEL

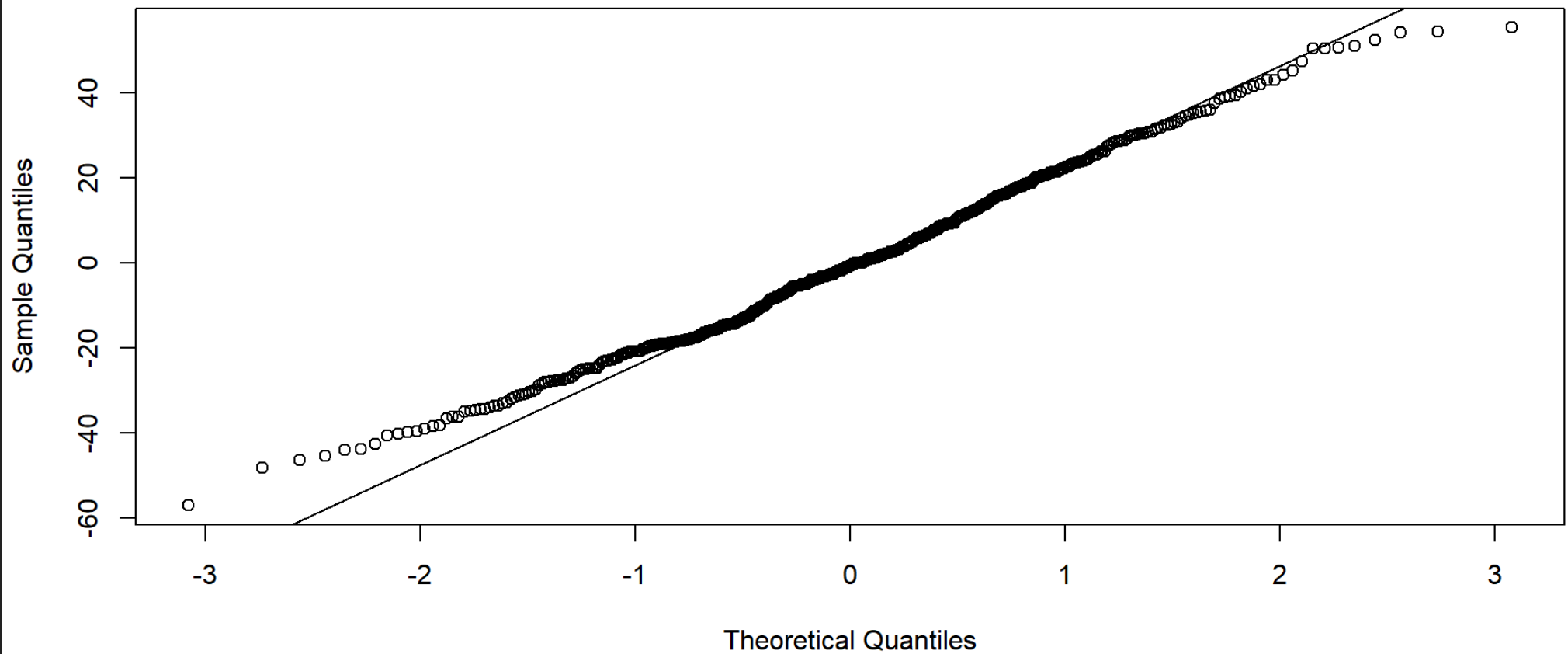
```
1 (prelim_plot <- ggplot(dragons, aes(x = bodyLength, y =  
2   geom_point() +  
3   geom_smooth(method = "lm")) +  
4   theme_classic())
```



ASSUMPTIONS CHECK

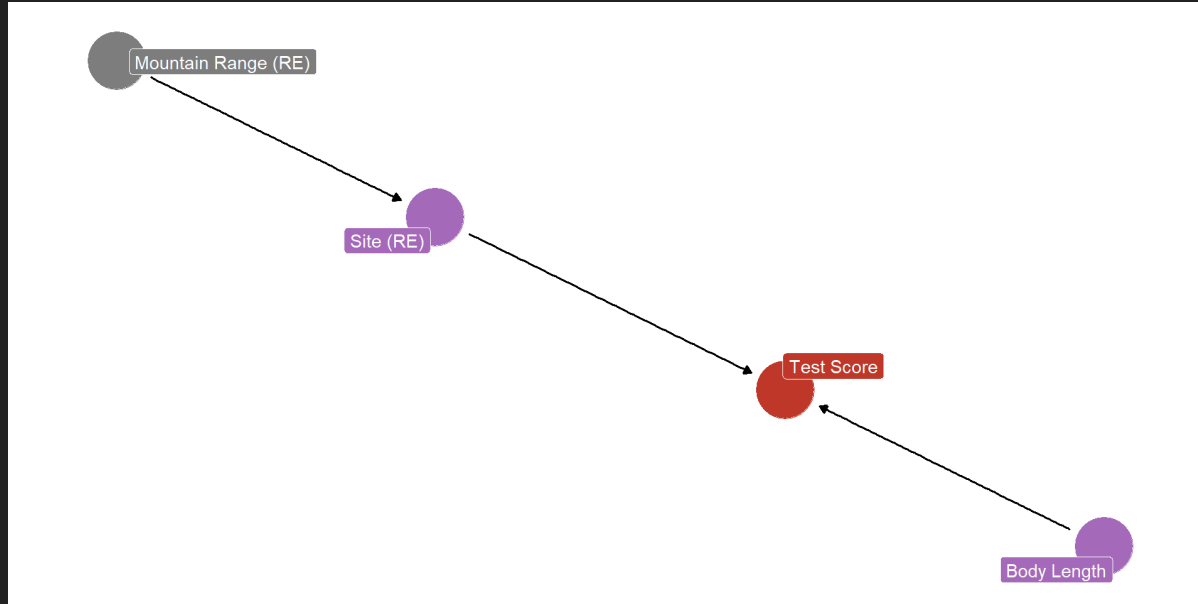
```
1 qqnorm(resid(basic.lm)); qqline(resid(basic.lm))
```

Normal Q-Q Plot



ASSUMPTIONS CHECK

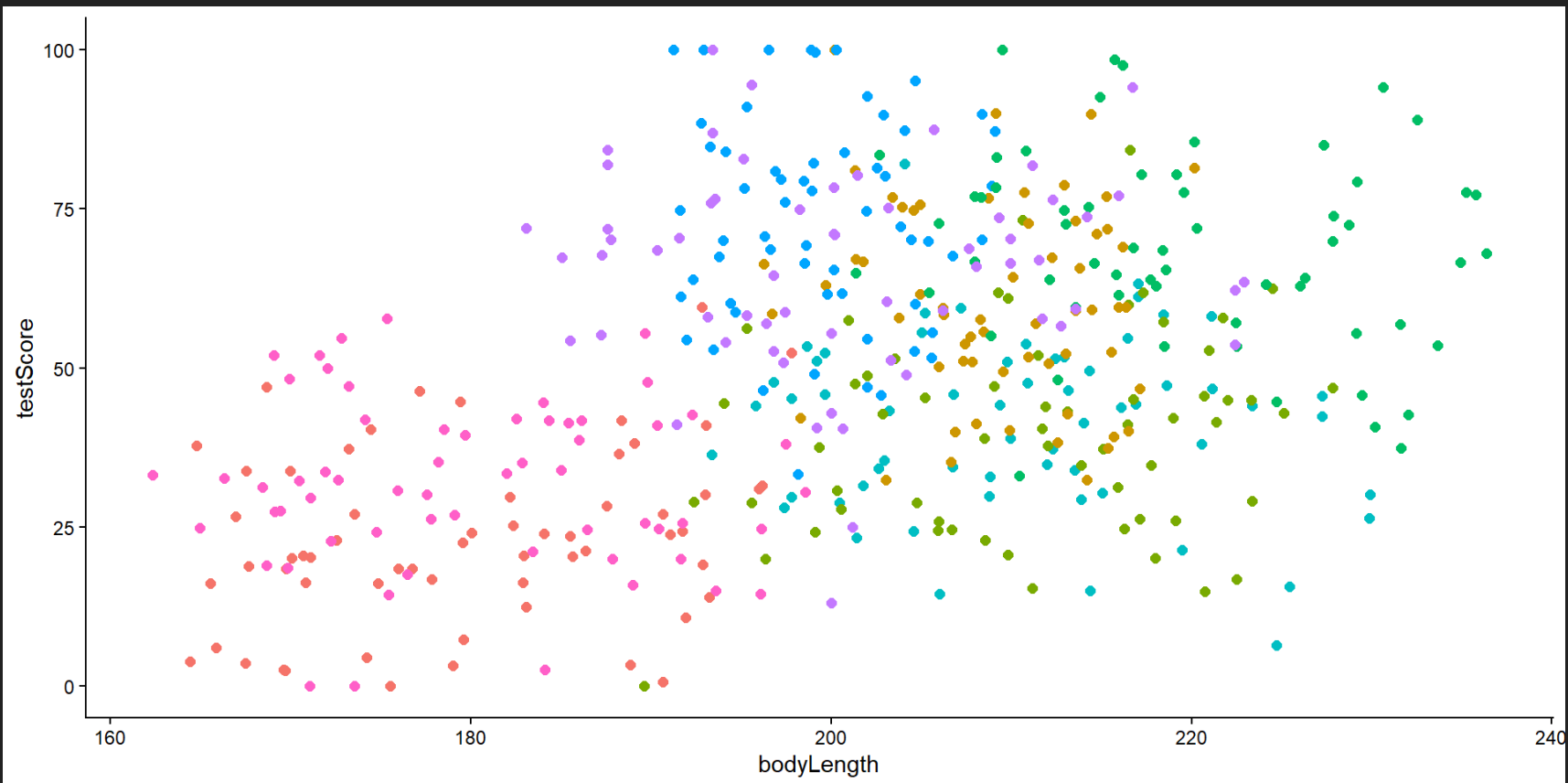
ARE OUR DATA INDEPENDENT?



- They are Hierarchical!
- Our research question:
 - Does a dragon's body length predict its test score, after accounting for variation across mountain ranges?

ASSUMPTIONS CHECK

```
1 (colour_plot <- ggplot(drakens, aes(x = bodyLength, y =  
2   geom_point(size = 2) +  
3   theme_classic() +  
4   theme(legend.position = "none"))
```



HOW TO IMPLEMENT MIXED MODELS IN R?

- Step 1: Model building
- Step 2: Model validation
- Step 3: Model interpretation
- Step 4: Model visualization

STEP 1: MODEL BUILDING

Hierarchical linear models fit a separate regression line for each and every cluster, then estimate what we call the *fixed slope*: the average slope between x and y across all clusters.

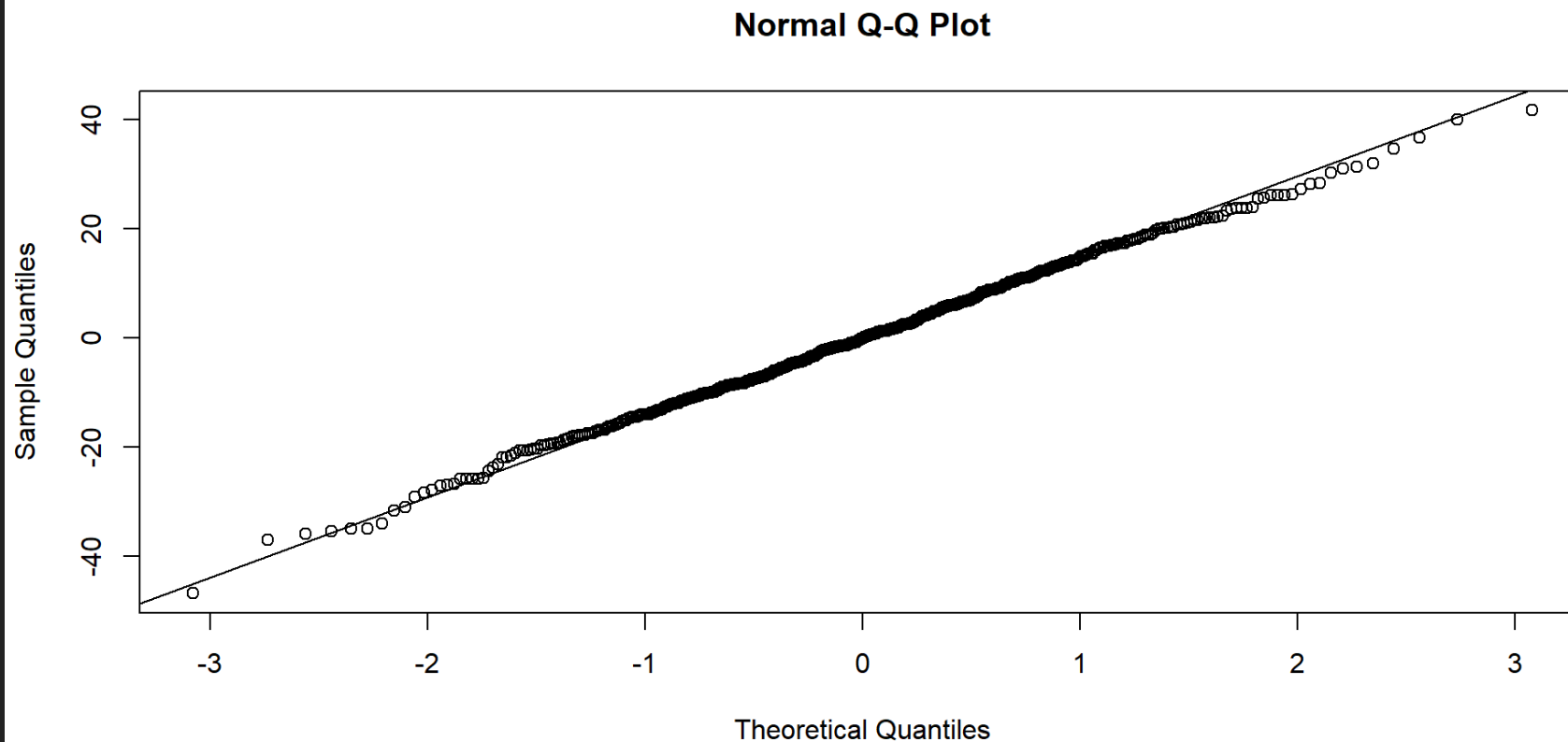
STEP 1: MODEL BUILDING DRAGONS

```
1 library(lme4) # "linear mixed model" function from lme4 p
2 mixed.lmer <- lmer(testScore ~ bodyLength2 +
3                     (1|mountainRange/site), #random eff
4                     data = dragons,
5                     REML = TRUE #estimation method other
6                     )
```

STEP 2: MODEL VALIDATION DRAGONS

Assumptions check

```
1 qqnorm(resid(mixed.lmer))  
2 qqline(resid(mixed.lmer))
```



STEP 3: MODEL INTERPRETATION DRAGONS

```
Linear mixed model fit by REML ['lmerMod']
Formula: testScore ~ bodyLength2 + (1 | mountainRange/site)
Data: dragons
```

```
REML criterion at convergence: 3970.4
```

```
Scaled residuals:
```

Min	1Q	Median	3Q	Max
-3.2425	-0.6752	-0.0117	0.6974	2.8812

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
site:mountainRange	(Intercept)	23.09	4.805
mountainRange	(Intercept)	327.56	18.099
Residual		208.58	14.442

Number of obs: 480, groups: site:mountainRange, 24; mountainRange, 8

```
Fixed effects:
```

	Estimate	Std. Error	t value
(Intercept)	50.386	6.507	7.743

```
1 library(MuMIn)
2 r.squaredGLMM(mixed.lmer)
```

	R2m	R2c
[1,]	0.001233396	0.6274854

STEP 4. VISUALIZATION

```
1 library(ggeffects)
2
3 # Extract the prediction data frame
4 pred.mm <- ggpredict(mixed.lmer, terms = c("bodyLength2")
5 head(pred.mm)
```

Predicted values of testScore

bodyLength2	Predicted	95% CI
-3	47.89	31.72, 64.07
-2	48.72	34.33, 63.12
-1	49.56	36.35, 62.76
0	50.39	37.60, 63.17
1	51.22	38.01, 64.42
2	52.05	37.66, 66.44

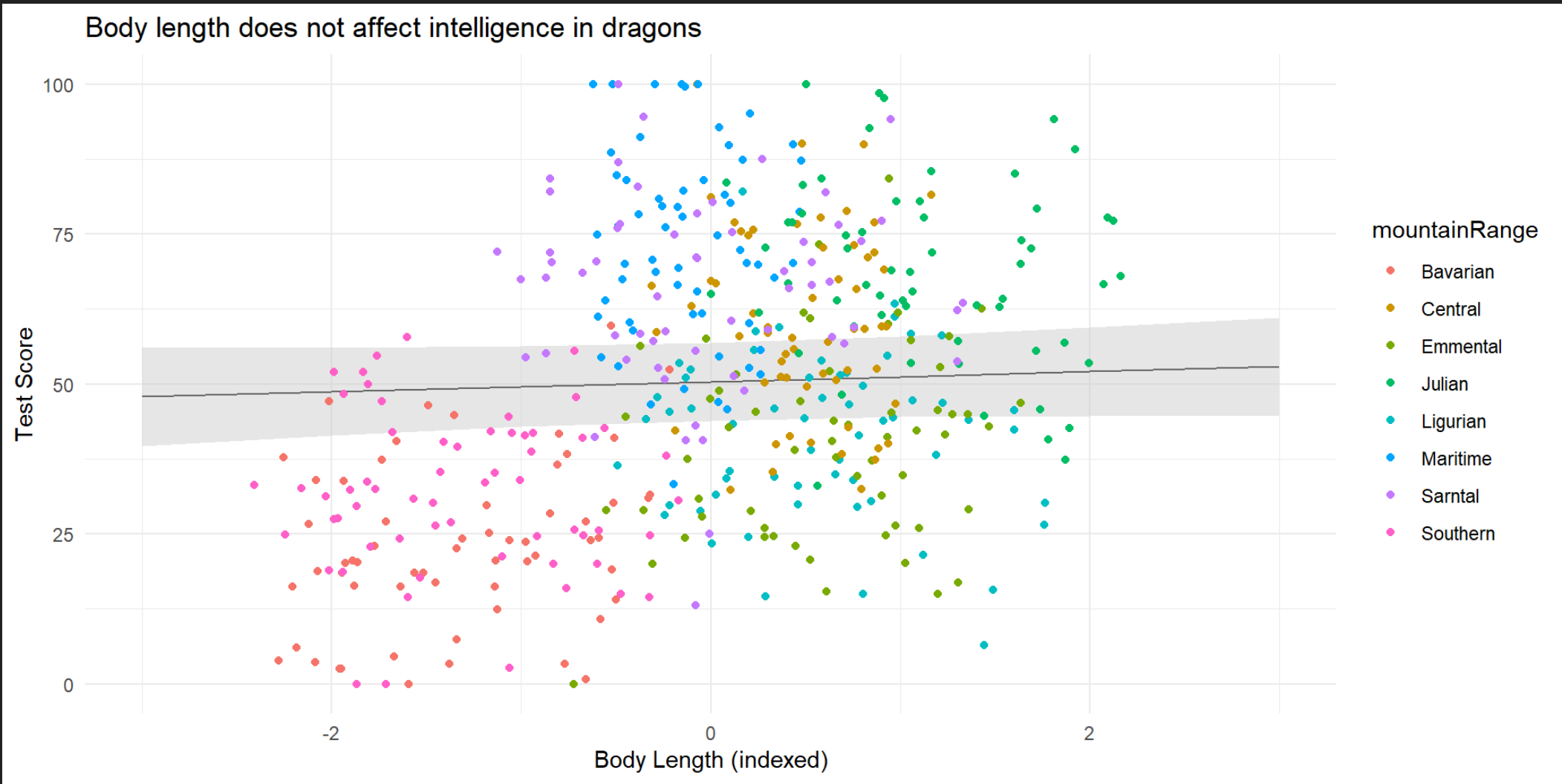
Adjusted for:

```
* site = 0 (population-level)
* mountainRange = 0 (population-level)
```

STEP 4. VISUALIZATION

```
1 # Plot the predictions
2 p<-ggplot(pred.mm) +
3   # slope
4   geom_line(aes(x = x, y = predicted)) +
5   # error band
6   geom_ribbon(
7     aes(
8       x = x,
9       ymin = predicted - std.error,
10      ymax = predicted + std.error
11    ),
12    fill = "lightgrey",
13    alpha = 0.5
14  ) +
15   # adding the raw data (scaled values)
16   geom_point(data = dragons,
```

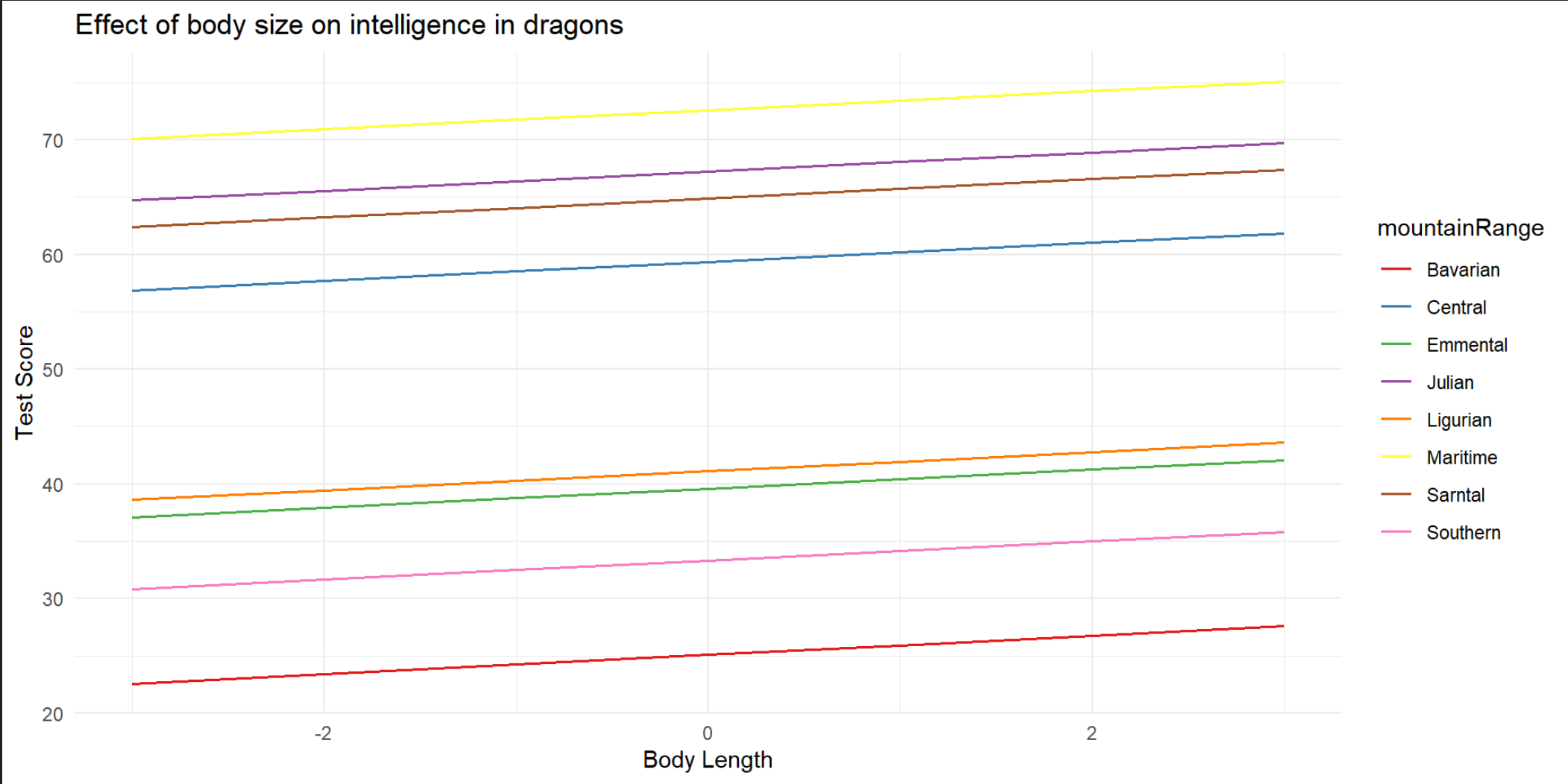
STEP 4. VISUALIZATION



STEP 4. VISUALIZATION

```
1 p<-
2   ggpredict(mixed.lmer, terms = c("bodyLength2", "mounta
3       type = "random") %>%
4   plot(show_ci = FALSE) +
5   labs(x = "Body Length", y = "Test Score",
6       title = "Effect of body size on intelligence in
7   theme_minimal()
```

STEP 4. VISUALIZATION



RANDOM INTERCEPT VS. RANDOM SLOPE

Random intercept ($1 | \text{group}$): each group gets its own baseline, but the effect of the predictor is the same across all groups.

Some mountain ranges have smarter dragons on average, but body length affects test score equally everywhere.

Random slope ($1 + x | \text{group}$): each group gets its own baseline **and** its own slope. The effect of the predictor is allowed to differ across groups.

In some mountain ranges body length predicts intelligence strongly; in others there is almost no relationship.

RANDOM INTERCEPT VS. RANDOM SLOPE

When to use which?

- Use a random intercept when you expect groups to differ in their average response but not in how they respond to your predictor.
- Use random slopes when you have biological reason to think the effect of your predictor genuinely varies across groups, and when you have enough data per group to estimate it.

RANDOM INTERCEPT VS. RANDOM SLOPE: THE SYNTAX

The **1** inside the brackets refers to the intercept. Adding a variable alongside it allows that variable's slope to vary too.

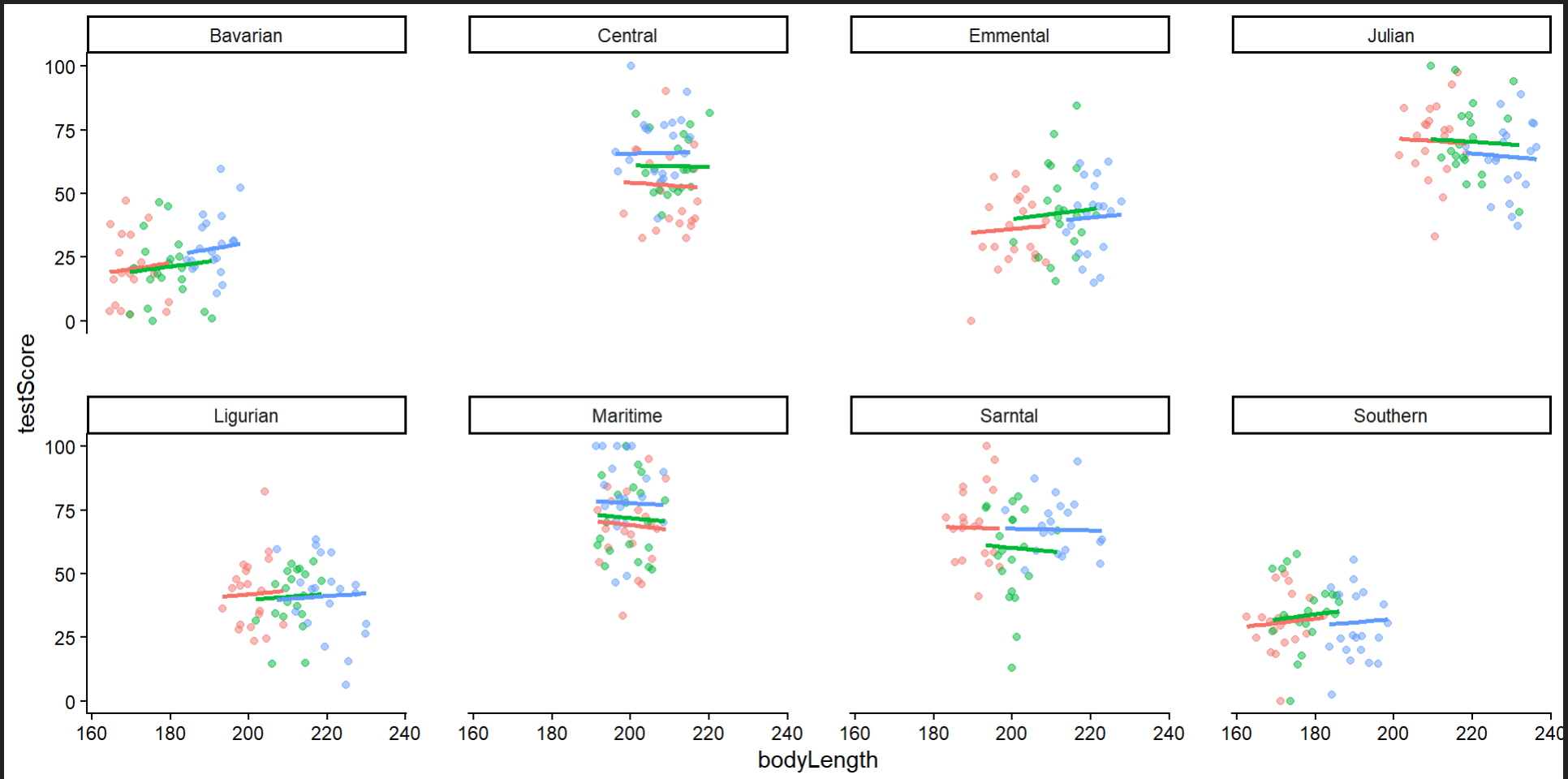
```
1 # Random intercept only: groups differ in baseline, same
2 lmer(y ~ x + (1|group))
3
4 # Random intercept + random slope: groups differ in base
5 lmer(y ~ x + (1 + x|group))
```

For the dragons: we allow both the baseline test score **and** the effect of body length to vary across mountain ranges and sites.

```
1 mixed.ranslope <- lmer(testScore ~ bodyLength2 +
2                       (1 + bodyLength2|mountainRange/
3                       data = dragons)
```

RANDOM SLOPES: WHAT IT LOOKS LIKE

Each panel is a mountain range. Each line is a site. Notice lines differ in both starting height (intercept) and steepness (slope).



NESTED VS. CROSSED

So far our random effects have been **nested**: each lower level belongs to exactly one higher level.

Site 1 in the Bavarian mountains is a different site from Site 1 in the Alps. They share a label but are completely separate. Sites are nested within mountain ranges.

Syntax: `(1|mountainRange/site)`

Crossed random effects arise when the same levels of one grouping variable appear across multiple levels of another.

The same **year** (e.g., 2022) occurs at every site simultaneously. Year is not inside any one site; it crosses all sites.

Syntax: `(1|site) + (1|year)` — two separate parallel terms.

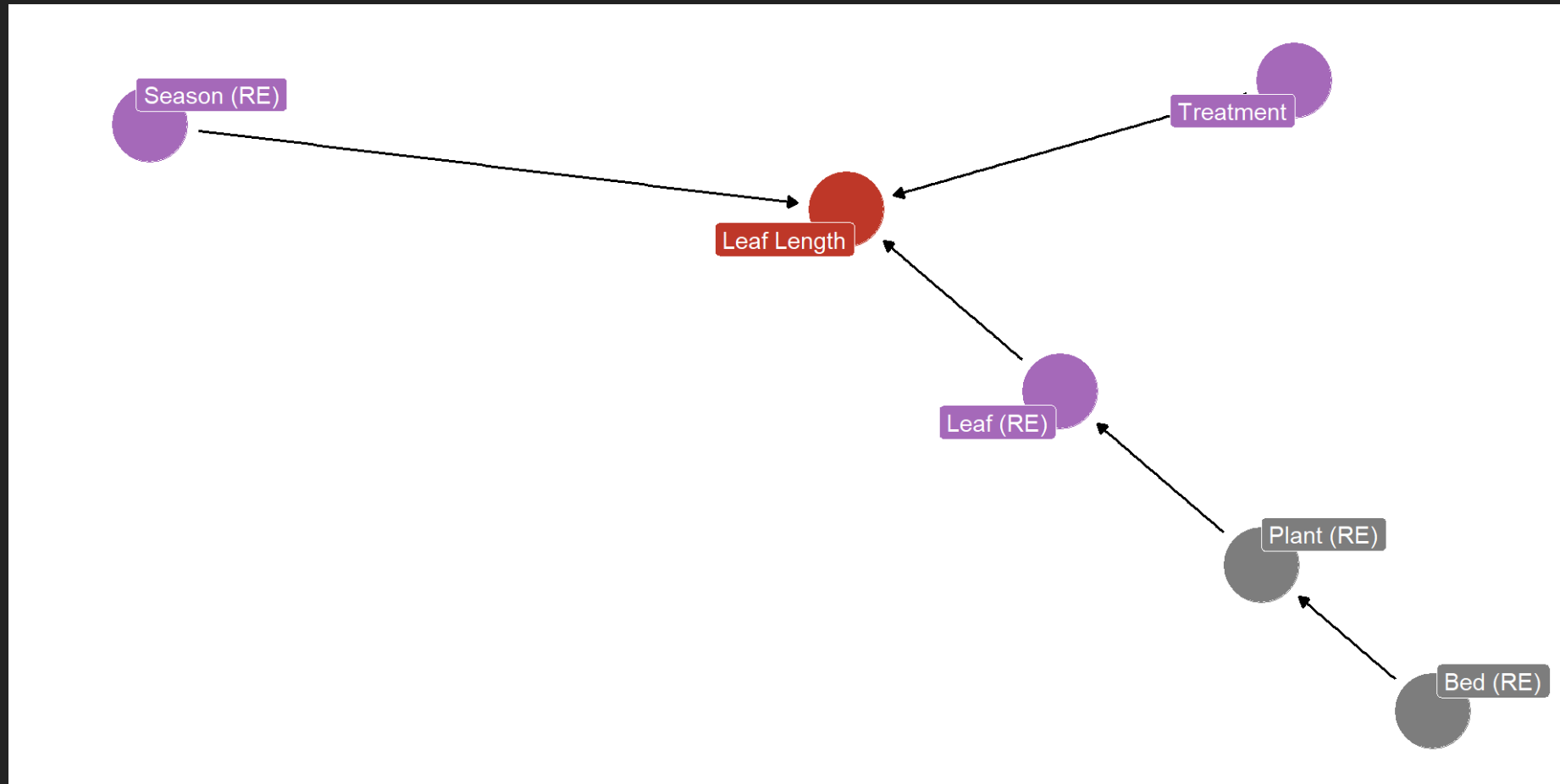
CROSSED EFFECTS: A PLANT EXAMPLE

Greenhouse experiment measuring leaf length under warming vs. control:

- 20 beds, each with 50 seedlings, each with 5 leaves measured: **nested** (Leaf inside Plant inside Bed).
- The experiment runs across 3 years and 4 seasons: **crossed** — every season occurs in every bed.
- A very hot summer hits in year 2, affecting all beds equally.

Season cannot go inside the Bed hierarchy because it is not unique to any bed. If you ignore it, the effect of that hot summer inflates your residual variance and distorts the treatment estimate.

CROSSED EFFECTS: THE SYNTAX



The nested hierarchy uses `/`. The crossed factor gets its own separate `+` term.

```
1 leafLength ~ treatment + (1|Bed/Plant/Leaf) + (1|Season)
```

ADDITIONAL RESOURCES

Popular libraries for (G)LMMs:

- Frequentist : [nlme](#), [lme4](#), [glmmTMB](#)
- Bayesian : [brms](#), [rstan](#), [rstanarm](#), [MCMCglmm](#)
- Nice visualization : [link](#)
- Most of the material comes from : [Coding Club](#)

BREAK



EXERCISE

EXERCISE

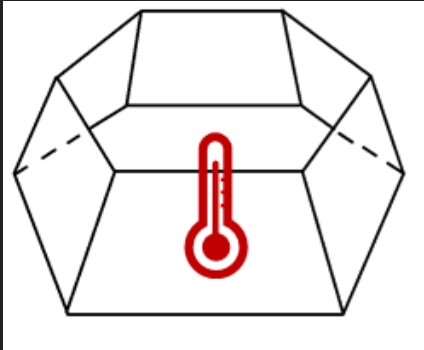
Now let's practice ...



ITEX

We will be using a dataset from the ITEX network.

- ITEX is a long-term warming experiment that uses standardized protocols to examine impacts of warming on Arctic ecosystems.
- Established in the 1990s - vegetation monitoring over three decades.
- Uses a simple method that is easy to establish in the field - open top chambers



ITEX



LET'S LOOK AT THE DATA

```
1 itex <- read_csv("Data/ITEX_diversity_data.csv")
```

```
1 head(itex)
```

```
# A tibble: 6 × 11
```

	...1	SITE	SUBSITE	PLOT	YEAR	TRTMT	Latitude	WarmQuarterTemp	SppRich
		<dbl>	<chr>	<chr>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
1	1	ALEXFIORD	ALEXFIORD:...	Cas....	2007	CTL	78.9	25.6	9
2	2	ALEXFIORD	ALEXFIORD:...	Cas....	2007	CTL	78.9	25.6	9
3	3	ALEXFIORD	ALEXFIORD:...	Cas....	2007	CTL	78.9	25.6	7
4	4	ALEXFIORD	ALEXFIORD:...	Cas....	2007	CTL	78.9	25.6	8
5	5	ALEXFIORD	ALEXFIORD:...	Cas....	2007	CTL	78.9	25.6	6
6	6	ALEXFIORD	ALEXFIORD:...	Cas....	2007	CTL	78.9	25.6	10

```
# i 2 more variables: `row_number()` <dbl>, PlotTemp <dbl>
```

```
1 length(unique(itex$SITE)) #this code tells you how many
```

```
[1] 24
```

```
1 unique(itex$SITE) #you can also do this and then it give
```

[1]	"ALEXFIORD"	"ANWR"	"ATQASUK"	"AUDKULUHEIDI"	"BARROW"
[6]	"BROOKS"	"BYLOT"	"DOVRE"	"ENDALEN"	"FAROE"
[11]	"GAVIA"	"KLUANE"	"KYTALYK"	"LATNJA"	"NIWOT"
[16]	"QHI"	"SADVENT"	"STEPSTONES"	"TAISETSU"	"THINGVELLIR"
[21]	"TIBET"	"TOOLIK"	"TORNGATS"	"VALBERCLA"	

EXERCISE 1: MODELING THE IMPACT OF CLIMATE CHANGE

Using the **ITEX diversity dataset**, think about a research question you could ask regarding the impact of one aspect of climate change on a variable of interest across different sites.

Explore your research question using modeling tools you learned in today's lecture.

EXERCISE 1 - YOUR TASKS

👉 Your Tasks:

1. **Model Planning:**

- What is your **response variable**?
- Which variables make sense as **fixed effects**?
- Which variable(s) should be treated as **random effects** (e.g., site, year)?

2. **Fit the Model:**

- Use `lmer()` to fit a model to the data and look at the output with `summary()`.

3. **Reflect:**

- Does the model structure reflect **how the data was collected**?

EXERCISE 2 (OPTIONAL)

Explore your research question **within sites**, as opposed to across sites as before.

Now that you've explored overall patterns, focus your model on how **climate change impacts your variable of interest, within each site.**

EXERCISE 2 (OPTIONAL) - YOUR TASKS

👉 Your Tasks:

1. • What changes when we ask about relationships *within sites* instead of *across* them?
 - Should you adjust the way **site** is treated in your model?
2. **Model Check:**
3. **Fit a New Model (if needed):**
 - Update your **lmer()** call to reflect this new focus.
 - Reinterpret the output

WHAT IS NEXT

- 28th of April 13:15am -17m *3401 Korallrevet*

